MATH2921 Year 2022 Exam EAQ. ①

Q1. Solution:

$$\begin{cases} u_{tt} - 3^2 u_{xx} = 0 & (c = 3) \\ u(x,0) = \cos(2x) & (f(x) = \cos(2x)) \\ u_t(x,0) = \sin x & (g(x) = \sin x) \end{cases}$$

D'Alembert's Solution

$$u(x,t) = \frac{1}{2}\Big(\cos(2(x+3t)) + \cos(2(x-3t))\Big) + \frac{1}{6}\int_{x-3t}^{x+3t} \sin(s)\,ds$$

$$= \frac{1}{2}\big(\cos(2x+6t) + \cos(2x-6t)\big) - \frac{1}{6}\big(\cos(x+3t) - \cos(x-3t)\big).$$

Q2. Solution: ②

$$\begin{cases} u_{xx} + u_{yy} = 0 & (x,y) \in (0,\pi) \times (0,2\pi) \\ u(x,0) = 0,\ u(x,2\pi) = \cos x & x \in [0,\pi] \\ u_x(0,y) = u_x(\pi,y) = 0 & y \in [0,2\pi] \end{cases}$$

Try (separated):

$$u(x,y) = X(x)\,Y(y)$$

Plug into PDE:

$$X''(x)Y(y) + X(x)Y''(y) = 0.$$

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = \lambda$$

$$\begin{cases} X''(x) - \lambda X(x) = 0 \\ Y''(y) + \lambda Y(y) = 0. \end{cases}$$

$u(x,0) = X(x)Y(0) = 0 \rightarrow Y(0)=0.$

$u_x(0,y) = X'(0)Y(y) = 0 \rightarrow X'(0)=0.$

$u_x(\pi,y) = X'(\pi)Y(y) = 0 \rightarrow X'(\pi)=0.$

$u(x,2\lambda) = X(x)Y(2\lambda) = \cos x$

$$\begin{cases} X''(x) - \lambda X(x) = 0 \\ X'(0) = X'(\pi) = 0 \end{cases}$$

Eigenvalue Problem ⓑ (with the opposite sign for λ):

Eigenvalue: $-\lambda_k = k^2$, $k = 0, 1, \dots$ $(\lambda_k = -k^2)$

Eigenfunction: $\cos(kx)$.

$k=0$: $Y_0''(y) = 0$, $Y_0(0) = 0$.

So $Y_0(y) = C_0 y$

$$\begin{cases} Y_k''(y) - k^2 Y_k(y) = 0 \\ Y_k(0) = 0 \end{cases} \quad k = 1, 2, \dots$$

$$Y_k(y) = C_{1,k} e^{ky} + C_{2,k} e^{-ky}$$

$0 = Y_k(0) = C_{1,k} + C_{2,k}$, $C_{2,k} = -C_{1,k} = -C_k$.

Formal general solution:

$$u(x,y) = C_0 y + \lambda \sum_{k=1}^{\infty} (C_k e^{ky} - C_k e^{-ky}) \cos(kx)$$

$$\cos x = u(x, 2\pi) = 2\pi C_0 + \lambda \sum_{k=1}^{\infty} (C_k e^{2k\pi} - C_k e^{-2k\pi}) \cos(kx)$$

Just Fourier Cosine Series:

$$C_1 e^{2\pi} - C_1 e^{-2\pi} = 1$$

$$C_k = 0, \quad k = 0, 2, 3, \dots$$

$$C_1 = \frac{1}{e^{2\pi} - e^{-2\pi}}.$$

So $u(x,y) = \dfrac{e^{y} - e^{-y}}{e^{2\pi} - e^{-2\pi}} \cdot \cos x.$

[Clearly an actual solution.]

Q3. Solution: $y'' - y' - 2 = 0.$

$y' = z.$ [Many method.]

$z' - z = 2$

$(e^{-x} z)' = 2e^{-x}$

$e^{-x} z = -2e^{-x} + C$

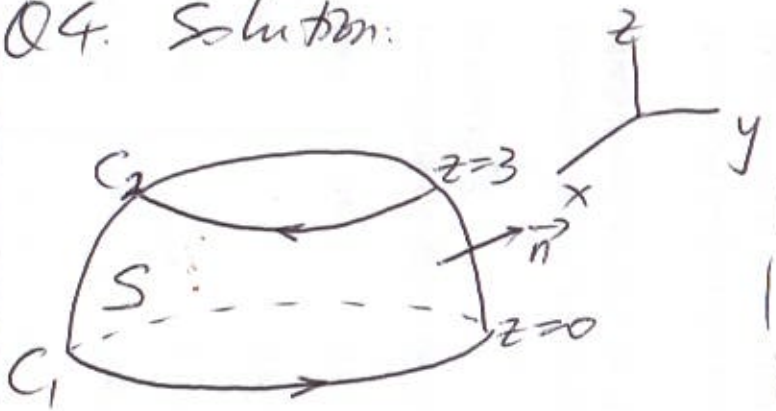
$z = -2 + ce^{x}$

$y' = -2 + ce^{x}$

$y = -2x + ce^{x} + D$

C, D constants

Q4. Solution:



$\iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, ds$

Stokes' Theorem

$= \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r}$

The boundary of S has 2 curves C_1 & C_2.

$$\begin{array}{lccc} & x(t) & y(t) & z(t) \\ C_1 : \vec{r}_1(t) = & (5\cos t, & 5\sin t, & 0) \end{array}$$

$t \in [0, 2\pi)$

counter-clockwise when viewed from positive z-axis.

(7)

$$\int_{C_1} \vec{F}\cdot d\vec{r}$$

$$= \int_0^{2\pi} (y(t), z(t), x(t))\cdot(x', y', z')\,dt$$

$$= \int_0^{2\pi} (5\sin t, 0, 5\cos t)\cdot(-5\sin t, 5\cos t, 0)\,dt$$

$$= \int_0^{2\pi} -25\sin^2 t\,dt$$

$$= -25\int_0^{2\pi} \tfrac{1}{2}(1-\cos 2t)\,dt$$

$$= -25\times\tfrac{1}{2}\times 2\pi$$

$$+ \cancel{\tfrac{25}{4}\sin 2t\Big|_0^{2\pi}}$$

$$= -25\pi$$

(8)

$$\int_{C_2} \vec{F}\cdot d\vec{r}$$

$C_2: \vec{r}_2(t) = (4\sin t, 4\cos t, 3)$

$t\in[0, 2\pi)$

clockwise when viewed from "above"

$$= \int_0^{2\pi} (y(t), z(t), x(t))\cdot(x', y', z')\,dt$$

$$= \int_0^{2\pi} (4\cos t, 3, 4\sin t)\cdot(4\cos t, -4\sin t, 0)\,dt$$

$$= \int_0^{2\pi} (16\cos^2 t - 12\sin t)\,dt$$

$$= 16\int_0^{2\pi} \tfrac{1}{2}(1+\cos 2t)\,dt + \cancel{12\cos t\Big|_0^{2\pi}}$$

$$= 16\times\tfrac{1}{2}\times 2\pi = 16\pi.$$

So $-25\pi + 16\pi = 9\pi$

Q5. Solution:

(9)

(i) $x' = Hy = -x + 2y$

$y' = Hx = 2x - y$

$$\vec{r}'(t) = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

(ii) $A = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda+1 & -2 \\ -2 & \lambda+1 \end{vmatrix}$$

$$= (\lambda+1)^2 - 4 = 0$$

$\lambda + 1 = \pm 2$

$\lambda = 1, -3$

(10)

$\lambda = 1$: $(I - A)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda = -3$: $(-3I - A)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix} \cdot \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$e^{tA} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-3t} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{tA} \begin{pmatrix} x(0) \\ y(0) \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-3t} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \boxed{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

$$= \begin{pmatrix} e^{t} & e^{-3t} \\ e^{t} & -e^{-3t} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} e^{t} \\ e^{t} \end{pmatrix}.$$

Q6. Proof / Solution:

(i) $d f(x, g(x)) / dx$

$$= f_x(x, g(x)) + f_y(x, g(x))\, g'(x) \quad (*)$$

$F(x, g(x)) = 0.$

$F_x + F_y\, g'(x) = 0.$ (☆)

$\nabla(f + \lambda F) = \vec{0}$ at $(x, y) = (a, b)$

So $f_x + \lambda F_x = 0$ (λ constant)

$f_y + \lambda F_y = 0$ at (a, b).

$g(a) = b$

Thus by (☆), $(*) = 0$.

(ii)

$$(1, g') \cdot D^2(f+\lambda F)\begin{pmatrix}1\\ g'\end{pmatrix}$$

$$= (1, g')\begin{pmatrix} f_{xx}+\lambda F_{xx} & f_{xy}+\lambda F_{xy} \\ f_{xy}+\lambda F_{xy} & f_{yy}+\lambda F_{yy}\end{pmatrix}\begin{pmatrix}1\\ g'\end{pmatrix}$$

$$= \underline{f_{xx}} + \lambda F_{xx} + g' f_{xy} + g'\lambda F_{xy}$$

$$+ g' f_{xy} + g'\lambda F_{xy} + \underline{\underline{g'^2 f_{yy}}}$$

$$+ g'^2 \lambda F_{yy}$$

So we need

$$f_y g'' = -\lambda F_y g''$$

at (a,b)

True at (a,b) since $\nabla(f+\lambda F) = \vec{0}$

$$d^2 f(x, g(x)) / dx^2$$

$$= \frac{d}{dx}(f_x + f_y g')$$

$$= \underline{f_{xx}} + f_{xy} g' + f_{xy} g' + \underline{\underline{f_{yy} g'^2}}$$

$$+ f_y g''$$

At (a, b), we need:

$$f_y g'' = \lambda F_{xx} + 2 g' \lambda F_{xy} + g'^2 \lambda F_{yy}$$

Recall $F(x, g(x)) = 0$.

$$F_x + F_y g' = 0$$

$\frac{d}{dx}$: $F_{xx} + 2F_{xy} g' + F_{yy}(g')^2 + F_y g'' = 0$